KAIZEN MATHS

Lesson starter

Advanced Matrices: Transformations and Eigenvalues

Matrix Equations and Systems

Name _____

S1

Solve each matrix equation AX = b.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, b = \begin{bmatrix} -11 \\ -13 \end{bmatrix}$

S2

Solve each triangular system by back substitution.

$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}, b = \begin{bmatrix} 10 \\ 8 \\ -8 \end{bmatrix}$

S3

Solve each matrix equation AX = B.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, B = \begin{bmatrix} 0 & 2 \\ -3 & -2 \end{bmatrix}$

S4

Solve each matrix equation AX = b.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, b = \begin{bmatrix} 16 \\ 23 \end{bmatrix}$

Show your working

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Exit ticket

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Solve each triangular system by back substitution.

$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 11 \\ 5 \\ 16 \end{bmatrix}$

EA2

Solve each matrix equation AX = B.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 13 \\ 1 & 14 \end{bmatrix}$

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Advanced Matrices: Transformations and Eigenvalues

Matrix Equations and Systems

Name _____

EB1

Solve each matrix equation AX = b.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$

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Advanced Matrices: Transformations and Eigenvalues

Matrix Equations and Systems

Name _____

EA1

Solve each triangular system by back substitution.

$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 11 \\ 5 \\ 16 \end{bmatrix}$

EA2

Solve each matrix equation AX = B.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 8 & 13 \\ 1 & 14 \end{bmatrix}$

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Advanced Matrices: Transformations and Eigenvalues

Matrix Equations and Systems

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S1

Solve each matrix equation AX = b.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $b = \begin{bmatrix} -11 \\ -13 \end{bmatrix}$

Answer:

$\begin{bmatrix} -4 \\ -3 \end{bmatrix}$

Working:

Solve AX = b.

Let $X = \begin{bmatrix} x \\ y \end{bmatrix}$.

$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -11 \\ -13 \end{bmatrix}$.

This gives the equations 2x + y = -11 and x + 3y = -13.

From 2x + y = -11, y = -11 - 2x.

Substitute into x + 3y = -13.

$x + 3(-11 - 2x) = -13$.

Solving gives x = -4.

Then y = -3.

So $X = \begin{bmatrix} -4 \\ -3 \end{bmatrix}$.

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S2

Solve each triangular system by back substitution.

$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, $b = \begin{bmatrix} 16 \\ 8 \\ -8 \end{bmatrix}$

Answer:

$\begin{bmatrix} 5 \\ 4 \\ -2 \end{bmatrix}$

Working:

Solve AX = b by back substitution.

Let $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

The third row gives 4z = -8.

So z = -2.

The second row gives 3y + 2z = 8.

Substitute z = -2: 3y - 4 = 8.

So y = 4.

The first row gives 2x + y - z = 16.

Substitute y = 4 and z = -2: 2x + 4 + 2 = 16.

So x = 5.

Therefore $X = \begin{bmatrix} 5 \\ 4 \\ -2 \end{bmatrix}$.

Teacher answers & working

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S3

Solve each matrix equation AX = B.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 2 \\ -3 & -2 \end{bmatrix}$

Answer:

$\begin{bmatrix} 1 & 2 \\ -2 & -2 \end{bmatrix}$

Working:

Solve AX = B.

This means solve a separate system for each column of X.

Let $X = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$.

Then AX = $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} p & q \\ r & s \end{bmatrix}$.

Equating the first column gives $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} p \\ r \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$.

Solving gives p = 1 and r = -2.

Equating the second column gives $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} q \\ s \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$.

Solving gives q = 2 and s = -2.

Therefore X = $\begin{bmatrix} 1 & 2 \\ -2 & -2 \end{bmatrix}$.

Teacher answers & working

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S4

Solve each matrix equation AX = b.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 16 \\ 23 \end{bmatrix}$

Answer:

$\begin{bmatrix} 5 \\ 6 \end{bmatrix}$

Working:

Solve AX = b.

Let $X = \begin{bmatrix} x \\ y \end{bmatrix}$.

$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 16 \\ 23 \end{bmatrix}$.

This gives the equations 2x + y = 16 and x + 3y = 23.

From 2x + y = 16, y = 16 - 2x.

Substitute into x + 3y = 23.

$x + 3(16 - 2x) = 23$.

Solving gives x = 5.

Then y = 6.

So $X = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$.

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EA1

Solve each triangular system by back substitution.

$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, $b = \begin{bmatrix} -11 \\ 5 \\ 16 \end{bmatrix}$

Answer:

$\begin{bmatrix} -3 \\ -1 \\ 4 \end{bmatrix}$

Working:

Solve AX = b by back substitution.

Let $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

The third row gives 4z = 16.

So $z = 4$.

The second row gives 3y + 2z = 5.

Substitute z = 4: 3y + 8 = 5.

So $y = -1$.

The first row gives 2x + y - z = -11.

Substitute y = -1 and z = 4: 2x - 1 - 4 = -11.

So $x = -3$.

Therefore X = $\begin{bmatrix} -3 \\ -1 \\ 4 \end{bmatrix}$.

Teacher answers & working

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EA2

Solve each matrix equation AX = B.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, B = \begin{bmatrix} 8 & 13 \\ 1 & 14 \end{bmatrix}$

Answer:

$\begin{bmatrix} 5 & 4 \\ -2 & 5 \end{bmatrix}$

Working:

Solve AX = B.

This means solve a separate system for each column of X.

Let $X = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$.

Then $AX = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix}$.

Equating the first column gives $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} p \\ r \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \end{bmatrix}$.

Solving gives p = 5 and r = -2.

Equating the second column gives $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} q \\ s \end{bmatrix} = \begin{bmatrix} 13 \\ 14 \end{bmatrix}$.

Solving gives q = 4 and s = 5.

Therefore $X = \begin{bmatrix} 5 & 4 \\ -2 & 5 \end{bmatrix}$.

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EB1

Solve each matrix equation AX = b.

$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$

Answer:

$\begin{bmatrix} 0 \\ 2 \end{bmatrix}$

Working:

Solve AX = b.

Let $X = \begin{bmatrix} x \\ y \end{bmatrix}$.

$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$.

This gives the equations 2x + y = 2 and x + 3y = 6.

From 2x + y = 2, y = 2 - 2x.

Substitute into x + 3y = 6.

$x + 3(2 - 2x) = 6$.

Solving gives x = 0.

Then y = 2.

So $X = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$.